



Creating Probability Distributions

Probabilities of all outcomes of a discrete random variable can be summarized in a table.

Creating Probability Distribution Tables

A study of a new treatment for high cholesterol undergoes testing. The results are summarized in this frequency table.

X	0–9	10–19	20–29	30–39	40–49
f	81	73	35	47	14

In this table, X stands for the number of points that the patient's cholesterol level decreased, and f stands for the number of patients. For instance, 73 patients had their cholesterol decrease between 10 and 19 points after the new treatment.

► **Remember** The letter X is often used to represent a discrete random variable. For example, when a coin is tossed, the variable X would represent the two outcomes: heads and tails. These outcomes are both discrete and random.

A patient wants to know the probability that the new treatment will lower his cholesterol from 20 to 29 points. Here, a **probability distribution table** would be useful because it shows the probabilities for all of the outcomes.

How to Create a Probability Distribution Table

Follow these steps to create a probability distribution table from a frequency table.

Step 1 Find the sum of the frequencies.

Step 2 Divide each frequency by the sum found in Step 1.

Step 3 Replace each frequency in the table with the ratio found in Step 2.

Step 4 Change the label f to P .

For the frequency table describing the clinical testing, the total number of frequencies is $81 + 73 + 35 + 47 + 14 = 250$. So 250 patients took part in the study. Here is the complete probability distribution table.

X	0–9	10–19	20–29	30–39	40–49
P	0.324	0.292	0.14	0.188	0.056

According to this table, the probability that the new treatment will lower a patient's cholesterol between 20 and 29 points is 0.14, or 14%.

► **Think About It** The sum of the probabilities in a probability distribution table always equals 1.

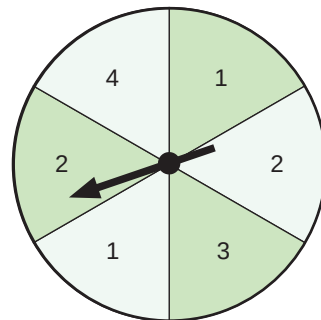
In a different study, patients are separated into four groups according to a spinner.

For the spinner, there are four different outcomes: Group 1, Group 2, Group 3, and Group 4. There are six sections overall.

Notice that some outcomes are more likely than others.

For instance, there are two sections for Groups 1 and 2, but only one section for Groups 3 and 4.

The probability distribution table shows that the chances of being in Group 1 or Group 2 are higher than the chances of being in Group 3 or Group 4.



X	1	2	3	4
P	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{6}$

Properties of Probability Distributions

There are two properties that every probability distribution table has.

Properties of Probability Distributions

Every probability distribution for a discrete random variable has the following:

- Each probability is a number between 0 and 1.
- The sum of all probabilities is equal to 1.

► **Think About It** Every probability distribution for a discrete random variable must satisfy both properties.

This probability distribution table violates the first property because each probability is not a number between 0 and 1. So it is not a probability distribution.

X	1	2	3	4	5
P	0.5	0.1	0.3	0.4	-0.3

This probability distribution table violates the second property because the sum of all probabilities is not equal to 1. So it is not a probability distribution.

X	5	10	15	20	25	30
P	0.34	0.5	0.101	0.32	0.34	0.122

Graphs of Probability Distributions

A probability distribution table and a histogram can be used to interpret the likelihood of an event or outcome.

The developer of a test preparation class wants to know the efficacy of the class in improving test scores. Students took a practice test on the first and last day of the class, and the difference in their scores was recorded. The results are shown in the table.

X	≤ 19	20–29	30–39	40–49	50–59	≥ 60
f	22	54	62	38	18	6

In the table, X stands for the number of points the score increased after taking the prep class, and f stands for the number of students.

A student wants to know the probability that the prep class will increase her test score by 30 to 39 points. Here, a probability distribution table would be useful.

Step 1 Find the sum of the frequencies. For this frequency table, the total number of frequencies is $22 + 54 + 62 + 38 + 18 + 6 = 200$. Two hundred students took part in the study.

Step 2 Divide each frequency by the total found in Step 1.

Step 3 Replace each frequency in the table with the ratio found in Step 2.

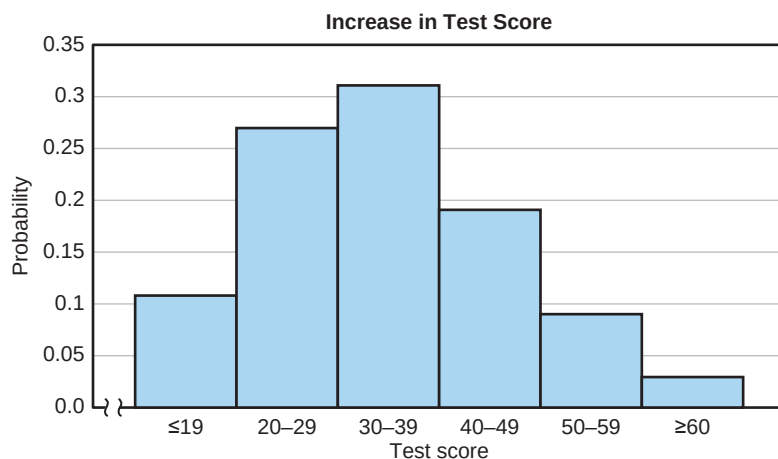
Step 4 Change the label f to P .

Here is the complete probability distribution table.

X	≤ 19	20–29	30–39	40–49	50–59	≥ 60
P	0.11	0.27	0.31	0.19	0.09	0.03

According to the table, the probability that the class will increase a student's score by 30 to 39 points is 0.31, or 31%.

These data can also be graphed in a probability distribution histogram.



The third bar of the histogram represents a 30- to 39-point increase in test scores. The height of this bar is at the value $P = 0.31$. The probability of a student increasing his or her score by 30 to 39 points is 0.31, or 31%.